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ECOLOGY

Model Estimates of Climate Change Impact on Habitats of Zonal Vegetation for the Plain Territories of Russia

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Abstract—Possible changes in the habitats of zonal phytocenoses for the plain territories of Russia under a 1°C increase in the annual mean global surface temperature are estimated by simulation with the atmosphere–ocean general circulation models ECHAM4/OPYC3 and HadCM3 and the intermediate-complexity climate model of the Institute of Atmospheric Physics for anthropogenic scenarios of greenhouse gas changes. The response of the phytocenotic habitats to possible climate changes is estimated from the changes in net primary production for the considered climatic scenarios. The obtained data allowed us to recognize the zonal phytocenoses most sensitive to climate changes.

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The structure and functioning of vegetation depend on many factors (climatic, biotic, edaphic, orographic, historic, etc.). However, the climatic impact on vegetation is so great that the impact of other factors is inferior within certain limits. As a result, vegetation largely remains constant under given climatic conditions despite a mosaic geological composition of the area (Kurnaev, 1973).

High latitudes over the Northern Hemisphere covering the bulk of Russia are experiencing considerable climatic changes. The annual mean global surface temperature increased by about 0.6°C during the 20th century, while the climate warming in Russia amounted to 0.9°C during the last century (*Climate Change*…, 2001). The analysis of the paleoclimatic data has shown that the rate of increase in the Northern Hemisphere temperature in the 20th century was much higher than the characteristic rate of its changes during the last two millennia (Mann and Jones, 2003). The data indicate a complex regional structure of the current climatic changes. A high spatial nonhomogeneity has been revealed in the surface temperature trends. In Russia, the significant trend in the annual mean surface temperature was observed in Central Siberia, Baikal Regions, Transbaikalia, Amur Regions, and Maritime Territory. Warming was more noticeable in winter and spring—4.7 and 2.9°C per century, respectively. In the second half of the 20th century, the annual and seasonal (except for winter) precipitation decreased in Russia (Gruza and Ran'kova, 2002). The most noticeable reduction in precipitation was observed in Northeastern Russia, whereas a weak tendency of precipitation growth was found in European Russia.

The inhomogeneity of climate changes predetermines a complex and ambiguous response of the vegetation cover to these changes. A trend to temperature increase and precipitation changes can cause serious changes in the vegetation structure and species composition. Climatic changes can have a significant impact on the timing of seasonal vegetation growth. An increased global surface temperature, changed precipitation amounts, an increased duration of the frost-free period, and other climatic characteristics may lead to reduction and fragmentation of the habitats of many plant species and to the occurrence of new existence conditions for individual plant communities and ecosystems (Kozharinov and Minin, 2001; Root et al., 2003).

In this work, possible changes in the habitats of potential zonal phytocenoses were evaluated for global climate warming in the first half of the 21st century and the zonal phytocenoses most sensitive to climatic changes were identified.

Vegetation habitats and climate. Principal distinctions between the habitats of different phytocenoses can be analyzed and their adaptation to particular climatic conditions can be compared using their phytoclimatic habitats. Phytoclimatic habitat is a graphical plot of a plant community habitat in coordinates of particular climatic factors. Phytoclimatic habitat at the regional level can be exemplified by the Grebenshchikov diagram (1986) plotting the habitats of natural phytocenoses in European Former Soviet Union in coordinates of annual precipitation and annual accumulated temperature above 10°C (Fig. 1a). This diagram allows us to comparatively analyze the adaptations of phytocenoses to heat and moisture conditions. Nearly

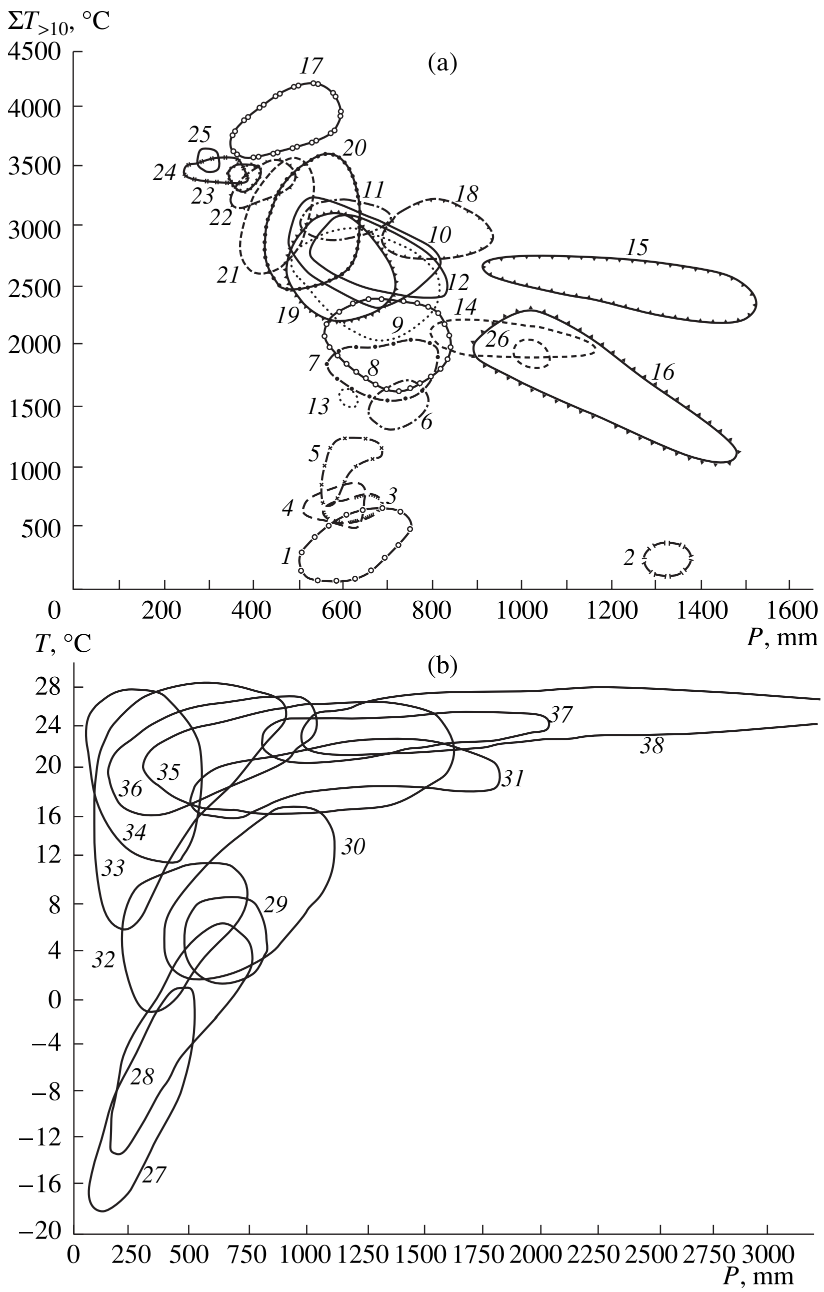

Fig. 1. Diagrams of phytoclimatic habitat distributions for the main natural ecosystems in European Former Soviet Union (a) and terrestrial ecosystems (b): *1*, plain tundra; *2*, mountain tundra; *3*, forest–tundra of *Betula tortuosa*; *4*, forest–tundra of *Picea obovata*; *5*, northern taiga spruce forest (*P. abies* and *P. obovata*); *6*, central taiga spruce forest (*P. abies* and *P. obovata*); *7*, southern taiga spruce forest (*P. abies* and *P. obovata*); *8*, subtaiga coniferous–broadleaf forest; *9*, oak–aspen and steppe pine forests; *10*, East European oak forest (*Quercus robur*); *11*, western oak forest of *Q. petraea*; *12*, hornbeam–oak mesophilic forest; *13*, Cisural coniferous–broadleaf forest (*Pinus sylvestris* and *Larix sucaczewii*); *14*, beech (and hornbeam–beech) forests at the foothills of Southeast Ukraine; *15*, fir (*Abies alba*) forest in southern Crimea; *16*, mountain spruce forest in Carpathians; *17*, juniper–oak (*Q. puescens*) forest in southern Crimea; *18*, pine (*Pinus stankewiczii*) mountain forest in southern Crimea; *19*, meadow steppe; *20*, forb–fescue–feather grass (true) steppe; *21*, fescue–feather grass true dry steppe; *22*, sagebrush–cereal dry steppe; *23*, sagebrush–cereal dry (desert) steppe; *24*, northern cereal–sagebrush semidesert; *25*, southern sagebrush–saltwort–cereal semidesert; *26*, steppe forest in Crimean mountain pasture; *27*, tundra; *28*, coniferous forest; *29*, mixed coniferous–broadleaf forest; *30*, broadleaf forest; *31*, tropical and subtropical evergreen forest; *32*, steppe, prairie, and pampas; *33*, semidesert; *34*, desert; *35*, savanna; *36*, tropical thin forest and shrubs; *37*, tropical forest deciduous in dry season; *38*, hylea; abscissa: annual precipitation (*P*); ordinate: (a) annual accumulated daily surface temperature above 10°C (10°C ($\sum T_{>10}$)) (Grebenshchikov, 1986) and (b) annual mean surface temperature (*T*) (Volobuev, 1947).

all habitats of forest phytocenoses occupy the thermal and precipitation space from 1500 to 3000°C and from 500 to 800 mm, respectively. In temperate latitudes, such space corresponds to the optimum for most plant communities, which is reflected by frequent overlapping of their phytoclimatic habitats. The habitats of phytocenoses adapted to extreme conditions lie below the above-mentioned group of phytoclimatic habitats; these include extremely cold (tundra and forest-tundra), cold and excessively wet conditions (e.g., mountain tundra in the Khibiny Massif), and waterlogged and moderate thermal conditions (e.g., dark coniferous forest in the Carpathian Mountains). The upper part of the diagram is occupied by phytocenoses adapted to dry conditions: various steppe types, semideserts, deserts, and sub-boreal dry forests (e.g., juniper pistachio oak forest and thin forest in the southern Crimea). Note that the extended shape of phytoclimatic habitats does not depend on the area occupied by the plant community. The narrowest habitat ranges in terms of temperature and moisture are observed in dry steppes and semideserts. The zonal changes in climatic conditions and spatial replacements of phytocenoses are continuously observed in the vast East European Plain. The location of phytocenotic habitats in this area often depends on the properties of soil-forming and bed rocks as well as the relief apart from climatic conditions (Grebenshchikov, 1986). This is reflected by overlapping of phytoclimatic habitats of the considered plant communities in the coordinates of specified climatic parameters.

At the global level, phytoclimatic habitats of many plant communities often overlap. Phytoclimatic habitats of terrestrial plant at the global level can be exemplified by the Volobuev diagram (1947). It reflects the arrangement pattern of the main terrestrial plant in the context of temperature and moisture changes (Fig. 1b). The bulk of the habitats of steppe and desert phytocenoses occupy the annual mean temperature and annual precipitation space from 12 to 18°C and below 650 mm, respectively. The habitats of forest phytocenoses in the temperate belt lie in the limits from –8 to 12°C and from 300 to 1000 mm, respectively. The habitats of subtropical and tropical phytocenoses largely occupy the space from 12 to 28°C and about 500 mm, respectively.

Thus, the phytoclimatic habitats of plant communities allow us to determine the limits of climatic factors permissible for individual phytocenoses or vegetation types. Due to the overlapping of phytoclimatic habitats for different phytocenoses, the phytocenotic habitat–climatic factor relationship is not one-to-one.

Simulation of the Vegetation Response to Climate Change

A widely used approach to study the changes in the spatial distribution of vegetation habitats is based on bioclimatic diagrams, i.e., empirical relationships between vegetation characteristics and some climatic parameters or their combination. The bioclimatic diagrams of Holdridge (1967), Lieth (1975), and Grigor'ev–Budyko (Budyko, 1977) are most frequently used in the models. The Holdridge diagram distinguishes plant communities by the following climatic variables: annual mean temperature, annual precipitation, and potential evaporation index. The Lieth diagram defines the limits of the main vegetation types from the annual mean temperature and annual precipitation. The Grigor'ev–Budyko diagram relates the geographic distribution of the main vegetation types to the surface radiation balance and radiation dryness index. The studies based on bioclimatic diagrams considering the relationship between the boundaries of plant communities and climatic parameters made it possible to evaluate the areas occupied by particular communities for various scenarios of climate changes (Belotelov et al., 1995; Kobak et al., 2002). Such studies usually assumed that the vegetation zones are transformed immediately after climate changes, i.e., they do not take into account that the actual shifts of phytocenotic habitats caused by global and regional climate changes are realized through successions and require tens and hundreds of years (Tishkov, 2005).

Gap models describing the plant community dynamics as responses of individual species are also used to evaluate the changes in spatial distribution of vegetation habitats (Urban et al., 1991; Chumachenko, 2004). Such models describe in detail biological features of plant growth, competition between plants, and the relationship between vegetation and climatic conditions. These models are used mainly to analyze the dynamics of forest vegetation. A detailed description of the physiological and ecological patterns in these models requires the availability of numerous quantitative empirical characteristics of the processes under study, which makes their application to simultaneously study the global changes in several vegetation types problematic.

The response of the global vegetation pattern to climate changes can also be assessed by using dynamic models. For example, a dynamic global vegetation model (Svirezhev and Zavalishin, 2003) makes it possible for a given climatic scenario to determine the pattern of the areas occupied by grass and forest vegetation at any time point and to estimate the time required for these vegetation types to reach an equilibrium state.

The impact of climatic parameters on vegetation regrowth can be evaluated in succession simulations. The mathematical formalism of discrete Markov chains is used in these simulations as a tool to describe the dynamics of plant communities in space and time (Logofet, 1999; Balzter, 2000). In a new type of Markov succession models (Logofet et al., 1997, 2005; Logofet, 1999), the probabilities of the successive transitions are not constant and depend on specific climate-sensitive changes in environmental factors. These environmental factors consequently depend on time due to

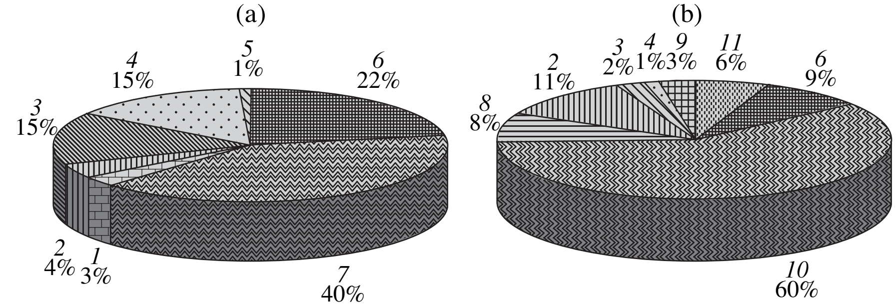

Fig. 2. Proportion of areas occupied by different forest types in European (a) and Asian (b) Russia: *1*, spruce–Siberian pine; *2*, spruce–Siberian pine–fir; *3*, broadleaf–coniferous; *4*, oak; *5*, limetree; *6*, pine; *7*, spruce; *8*, larch–pine; *9*, aspen–birch; *10*, larch; *11*, Japanese stone pine.

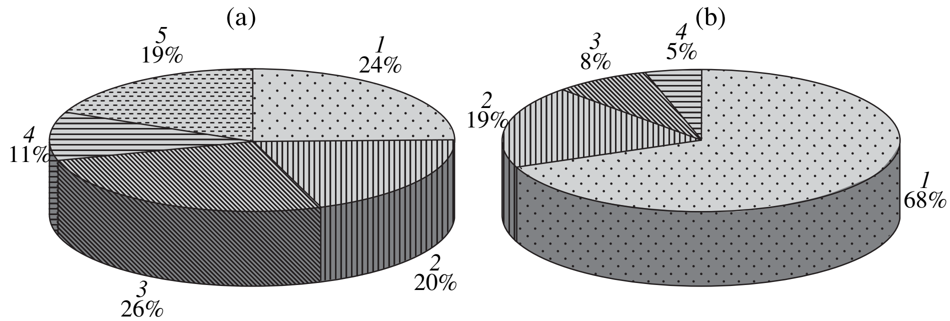

Fig. 3. Proportion of areas occupied by grass and tundra vegetation in European (a) and Asian (b) Russia: *1*, tundra; *2*, meadow steppe; *3*, true steppe; *4*, dry steppe; *5*, semidesert.

the accepted scenarios of climatic changes. Models of this type can describe vegetation changes during succession as a function of climate-dependent environmental factors and anthropogenic impact on ecosystems. Note that application of these models to evaluate possible changes in the vegetation habitats of certain ecosystems meets complications related to the definition of the succession pattern, i.e., a set of succession stages and transitions between them possible within a specified time interval.

The changes in the zonal structure of vegetation can be evaluated using the method of paleoanalogs. Application of this method allowed Velichko (2002) to evaluate possible changes in vegetation state in Russia during the 21st century resulting from anthropogenic climatic changes. However, the method of paleoanalogs generates the reconstructions reflecting an equilibrium between the zonal vegetation and climate, while the changes expected in the nearest decades will not be able to reach such equilibrium states.

MATERIALS AND METHODS

According to the map of restored vegetation (Bazilevich, 1993), 69 zonal phytocenoses are recognized on plain areas in Russia in the second half of the 20th century. We have chosen the following zonal vegetation types to present the results: forest (spruce, pine, spruce–Siberian pine, spruce–Siberian pine–fir, larch, larch–pine, broadleaf–coniferous, oak, limetree, aspen–birch, and Japanese stone pine), steppe (meadow, true steppe, and dry steppe), tundra, and semidesert. The proportions between the areas occupied by these vegetation types in European and Asian Russia are given in Figs. 2 and 3. The spatial pattern of the considered phytocenoses in Russia depends on specific intervals of climatic parameters. The variation ranges of some climatic parameters for the habitats of considered phytocenoses are given in Table 1. Note that the data in this table indicate the absence of one-to-one relationships between phytocenotic habitat and climatic factors for the zonal phytocenoses in Russia.

To estimate the phytocenotic habitats response to possible climate changes, we used the output of the atmosphere–ocean general circulation climate models ECHAM4/OPYC3 (Max-Planck-Institute for Meteorology, Germany) (Roeckner et al., 1996), HadCM3 (Met Office Hadley Centre, United Kingdom) (Collins et al., 2001) and the intermediate-complexity climate model developed at the Institute of Atmospheric Physics, Russian Academy of Sciences (IAP RAS CM) (Petoukhov et al., 1998; Mokhov et al., 2005).

The regional changes in the vegetation habitats in Russia were estimated under anthropogenic warming corresponding to a 1°C increase in the annual mean

Table 1. The limits of evaporation (E), radiation balance (R), annual precipitation (P), and mean temperatures in January (T_w) and July (T_s) for the habitats of zonal vegetation in Russia

Vegetation type	E, mm/year	R, GJ/(m^2 year)	P, mm/year	T_w, °C	T_s, °C
			European Russia		
Tundra	45.7–181.2	0.5–1.0	400.0–820.0	–22.0…–6.0	3.8–13.0
Spruce forest	135.7–400.0	0.8–1.4	580.0–820.0	–22.0…–6.6	10.0–18.8
Spruce–Siberian pine forest	194.4–275.0	1.1–1.2	783.3–870.0	–19.3…–15.7	13.3–17.1
Spruce–Siberian pine–fir Forest	277.8–311.4	1.2–1.4	675.0–787.5	–16.1…–6.7	16.5–18.9
Pine forest	128.6–391.7	0.8–1.5	590.0–856.0	–18.8…–6.4	8.0–19.2
Mixed forest	294.4–450.0	1.3–1.7	630.0–870.0	–16.3…–6.2	16.7–19.6
Oak Forest	300.0–465.0	1.4–2.1	500.0–870.0	–15.1…0.0	17.9–22.5
Limetree forest	300.0–320.0	1.5–1.7	615.0–800.0	–16.4…–15.5	17.3–19.0
Meadow steppe	300.0–455.6	1.4–2.1	540.0–787.5	–16.1…–2.5	17.4–22.5
True steppe	266.7–455.0	1.6–2.1	450.0–760.0	–16.0…–2.1	19.8–23.9
Dry steppe	160.7–455.0	1.7–2.1	350.0–650.0	–15.4…–1.0	22.0–24.5
Semideserts	156.7–455.0	1.7–2.1	216.7–650.0	–12.6…–2.0	22.6–25.5
			Siberia and Far East		
Tundra	35.7–157.1	0.3–1.2	320.0–800.0	–39.0…–16.0	–0.4–14.2
Spruce–Siberian pine–fir Forest	159.1–355.0	1.0–1.4	500.0–870.0	–33.6…–16.6	13.6–19.2
Pine forest	175.0–342.9	1.0–1.8	357.1–800.0	–26.0…–16.0	13.3–20.8
Larch forest	48.6–322.7	0.6 –1.0	320.0–870.0	–44.0…–16.0	8.0–18.2
Larch–pine forest	162.5–256.7	1.1–1.6	364.4–666.7	–38.6…–18.0	12.0–19.5
Japanese stone pine forest	111.8–246.4	0.9–1.5	360.0–1624.0	–40.0…–4.2	7.0–13.1
Mixed forest	302.1–341.3	1.6–2.0	590.0–823.3	–28.0…–12.0	16.3–19.3
Aspen–birch forest	242.0–350.0	1.3–1.5	433.3–800.0	–20.6…–16.6	17.5–19.5
Oak Forest	306.3–341.3	1.7–1.9	590.0–814.0	–27.8…–12.0	16.9–22.7
Meadow steppe	226.5–340.0	1.3–1.6	366.7–852.5	–24.5…–16.4	16.0–20.0
True steppe	246.1–314.3	1.4–1.8	333.3–595.0	–28.0…–16.3	16.4–21.5
Dry steppe	225.0–300.0	1.5–1.8	325.0–600.0	–29.0…–16.1	16.4–21.9

Note: Data for the second half of the 20th century (“Mirovoi vodnyi...,” 1974; “Klimat Rossii,” 2001).

global surface temperature, which is expected by the considered climatic models in late 2030s–early 2050s. According to these models, the spatial changes in the climatic conditions in Russia during warming are highly heterogeneous. The maximum warming is expected in winter, while less significant changes in the surface temperature are probable in summer (Table 2). The predicted changes in annual precipitation considerably vary. The IAP RAS CM scenario assumes an increase in precipitation throughout Russia except certain northeastern regions. The greatest increase in presentation is predicted by the model in semidesert regions in southern European Russia and vast areas in West and Central Siberia occupied by southern taiga larch–pine and larch forests. The ECHAM4/OPYC3 and HadCM3 scenarios assume decreased precipitation in tundra, steppe, and semidesert areas of European Russia. According to the ECHAM4/OPYC3 scenario, the greatest increase in precipitation is expected at the areas of southern taiga dark coniferous and aspen–birch forests in West Siberia and in broadleaf–coniferous and oak forests in Manchuria. According to the HadCM3 scenario, a considerable increase in precipitation should be expected at vast areas of central and southern taiga in West and Central Siberia.

Evaluation of changes in phytocenotic habitats for regrowth vegetation during climate warming based on changes in climatic conditions is problematic since the relationship between phytocenotic habitats and climatic conditions is not one-to-one. The changes in the net primary production (NPP) of phytocenoses during the considered climatic scenarios were used to solve this problem.

NPP is one of the main quantitative characteristics of phytocenoses. It depends on the organic matter increment within a specified time interval (commonly, a year) and a unit area for the overground and underground parts of a plant community. In this work, we

Table 2. The mean changes in evaporation (ΔE), radiation balance (ΔR), annual precipitation (ΔP), and mean temperatures in January (ΔT_w and July (ΔT_s) according to the ECHAM4/OPYC3, IAP RAS CM, and HadCM3 scenarios for the habitats of zonal vegetation in Russia

Vegetation type	ΔE, mm/year			ΔR, GJ/(m^2 year)			ΔP, mm/year			ΔT_w, °C			ΔT_s, °C		
	ECHAM	IAP	HadCM	ECHAM	IAP	HadCM	ECHAM	IAP	HadCM	ECHAM	IAP	HadCM	ECHAM	IAP	HadCM
European Russia															
Tundra	22.9	9.5	4.9	0.0	0.0	0.0	–8.1	51.7	–50.8	5.7	4.7	–0.3	1.0	1.3	0.6
Spruce forest	27.2	21.6	–15.8	0.2	0.2	0.2	38.5	41.1	–5.2	3.4	2.5	0.3	1.2	1.5	2.0
Spruce–Siberian pine forest	31.8	21.7	8.0	0.2	0.2	0.2	71.1	44.2	18.5	3.7	2.5	–1.0	1.4	1.6	2.0
Spruce–Siberian pine–fir Forest	22.7	24.2	–16.3	0.2	0.2	0.2	88.7	44.9	49.1	2.6	2.5	–0.1	1.5	1.8	2.4
Pine forest	26.4	21.9	–12.8	0.2	0.2	0.2	37.7	41.8	17.0	3.5	2.5	0.5	1.1	1.5	2.0
Mixed forest	21.5	23.6	–10.1	0.2	0.2	0.2	33.9	46.9	13.1	2.0	2.6	0.9	1.3	1.6	2.6
Oak Forest	21.0	22.3	26.9	0.1	0.1	0.2	51.0	55.0	–3.6	1.5	2.8	0.7	1.5	1.6	3.2
Limetree forest	11.1	21.2	–2.1	0.2	0.2	0.2	53.0	63.2	62.2	2.3	2.9	0.3	2.1	1.7	2.7
Meadow steppe	14.7	23.5	32.6	0.1	0.1	0.2	55.7	55.3	13.5	1.5	2.8	0.7	1.7	1.5	3.3
True steppe	16.1	29.9	66.9	0.2	0.1	0.2	35.6	59.2	29.5	1.3	2.6	0.8	1.8	1.4	2.9
Dry steppe	12.7	32.1	94.5	0.2	0.2	0.2	–33.3	68.9	–7.9	1.4	2.7	0.9	2.2	1.5	2.5
Semideserts	16.4	32.2	103.5	0.1	0.1	0.1	–54.0	78.7	–6.8	1.5	2.7	0.9	2.4	1.5	2.3
Siberia and Far East															
Tundra	16.5	27.2	4.4	0.0	0.0	0.0	40.6	47.0	54.6	5.5	6.1	2.1	1.7	1.3	0.8
Spruce–Siberian pine–fir Forest	27.4	21.2	–101.7	0.3	0.2	0.1	56.2	66.7	115.6	4.7	2.8	0.6	2.1	1.8	1.6
Pine forest	26.4	21.9	1.8	0.2	0.2	0.1	58.7	66.1	80.3	4.7	2.8	–0.1	1.8	1.6	1.7
Larch forest	22.0	20.4	–11.0	0.1	0.1	0.1	21.2	56.2	129.8	5.5	3.0	2.1	1.7	1.8	1.1
Larch–pine forest	26.1	28.9	–67.3	0.2	0.2	0.1	29.2	79.7	185.0	4.2	2.9	1.1	1.9	2.1	1.6
Japanese stone pine forest	9.2	33.9	84.8	0.1	0.1	0.1	–49.8	25.1	78.3	1.9	0.9	3.1	1.8	1.3	1.6
Mixed forest	19.7	46.9	206.0	0.1	0.1	0.2	73.1	33.0	147.8	3.4	1.5	2.0	0.8	1.0	2.1
Aspen–birch forest	23.6	24.6	–50.3	0.3	0.2	0.2	83.9	66.2	130.9	3.3	2.8	0.5	2.3	1.8	2.0
Oak Forest	12.5	50.7	208.2	0.1	0.1	0.2	99.9	37.8	146.0	3.2	1.5	1.7	0.8	1.0	1.9
Meadow steppe	16.7	26.6	–37.2	0.2	0.2	0.2	46.9	64.7	133.3	2.9	2.9	1.0	2.4	1.8	2.0
True steppe	8.5	25.6	94.5	0.2	0.1	0.2	69.1	63.4	148.8	2.9	2.8	1.3	2.3	1.5	2.0
Dry steppe	8.2	26.8	124.7	0.2	0.1	0.1	43.0	63.6	142.5	2.7	2.6	1.5	2.1	1.3	1.9

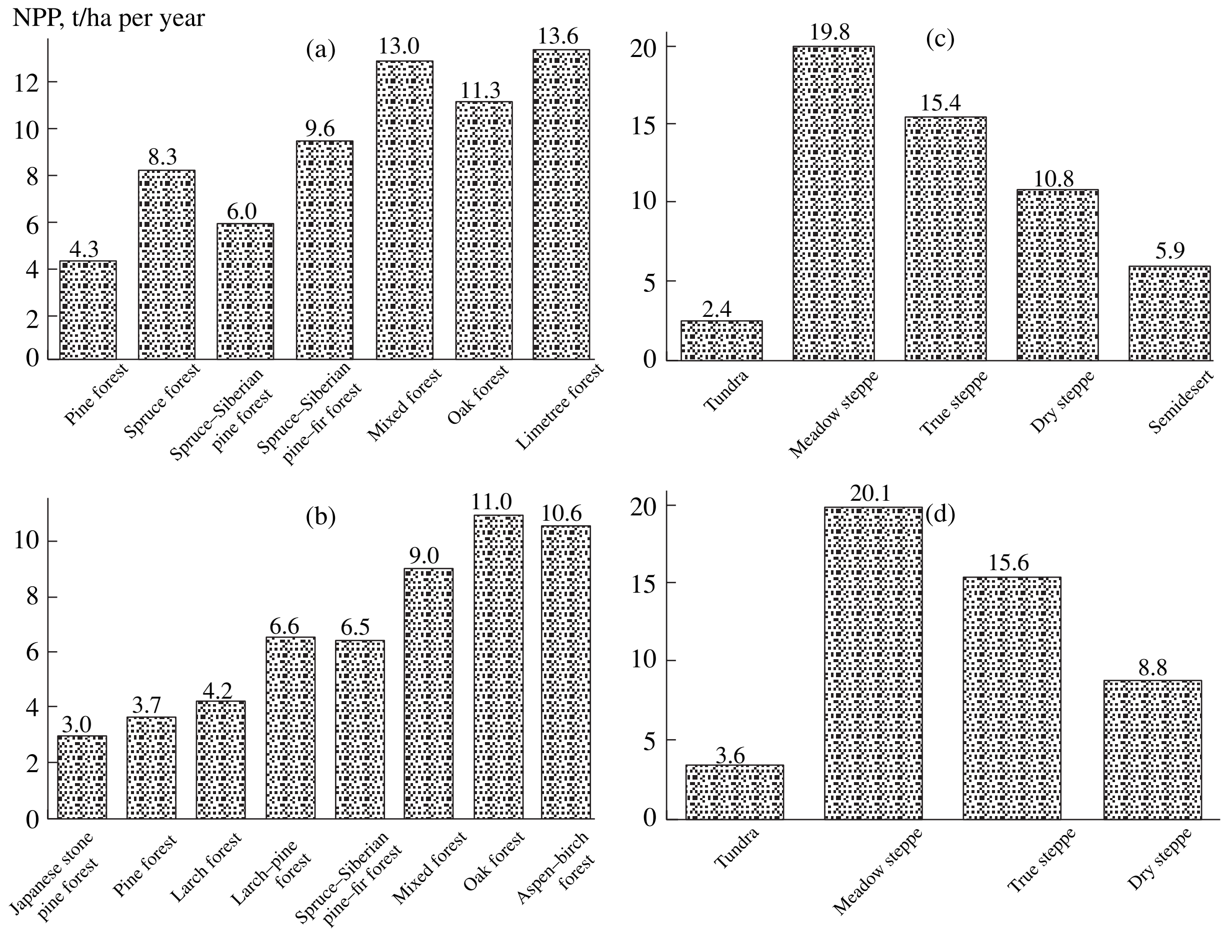

Fig. 4. Net primary production (NPP) of forest vegetation types in European (a) and Asian (b) Russia and grass vegetation types in European (c) and Asian (d) Russia (Bazilevich, 1993).

used the database of experimental NPP for Russian phytocenoses (Bazilevich, 1993). Forest phytocenoses in European Russia (Fig. 4a) feature NPP range from 4.3 t/ha per year (northern taiga pine forests) to 13.6 t/ha per year (Urals limetree forests). NPP of Siberian and Far Eastern forests (Fig. 4b) varies from 3 /ha per year (Japanese stone pine and alder forests) to 11 t/ha per year (Far Eastern oak forests). Among nonforest phytocenoses, the lowest NPP is observed in tundra—about 2.4 and 3.6 t/ha per year in the European (Fig. 4c) and Asian (Fig. 4d) Russia, respectively. The highest NPP is observed in meadow steppe—about 20 t/ha per year.

Complex nonlinear relationships exist between some climatic parameters and NPP (Varlygin and Bazilevic, 1992; Golubyatnikov and Denisenko, 2001a). In particular, NPP of Russian vegetation can be presented as a continuous single-valued function $f(R, E)$, where R is the radiation balance and E is the total evaporation (Fig. 5). This relationship underlies the model predicting the response of terrestrial NPP to climatic changes (Golubyatnikov and Denisenko, 2001b; Golubyatnikov et al., 2005).

Let us define the following rules of phytocenotic changes as a function of its NPP:

—if the predicted decrease in NPP of the phytocenoses is equal or more than NPP for the second half the 20th century, this phytocenosis disappears; if such decrease is observed only at a part of the habitat, this phytocenosis disappears only there;

—if predicted increase in NPP of a phytocenosis is at least 1.5 times that in the second half the 20th century, this phytocenosis regularly changes in time, which can be realized by succession only; if this increase is observed only at a part of the habitat, this phytocenosis undergoes succession only there.

Note that successions continuously go in phytocenoses due to various natural and anthropogenic factors. This work only considers successions induced by climatic factors.

To determine the response of the habitats of zonal phytocenoses to climate changes, we used predicted changes in NPP and accepted rules of phytocenotic changes as a function of its NPP. We assume a phytocenosis sensitive to climatic factors if more than 30% of the total area occupied by it undergoes changes.

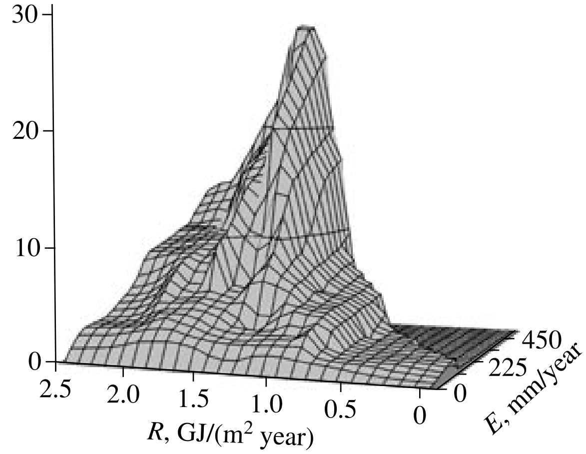

Fig. 5. Dependence of the net primary production (NPP) on radiation balance (R) and total evaporation (E) in Russian regions.

RESULTS AND DISCUSSION

Possible changes in NPP of the plain areas in Russia under a 1°C increase in the annual mean global surface temperature were calculated by using the climate models ECHAM4/OPYC3, HadCM3, and IAP RAS CM (Golubyatnikov et al., 2005). The obtained results indicate that the climate changes will be positive for forest phytocenoses in Russia. According to the HadCM3 scenario, an insignificant NPP decrease is expected in larch–pine and spruce–Siberian pine–fir forests in Siberia and Far East as well as in mixed and oak forests in European Russia (Fig. 6a). A similar prediction for NPP of oak forests in European Russia is given by the IAP RAS CM scenario. According to the HadCM3, a significant NPP increase is probable in mixed forests in Siberia and oak forests in the Far East (Fig. 6b). The IAP RAS CM model predicts a considerable NPP increase for limetree and oak–limetree submountain forests in southern Urals. No considerable NPP changes are expected for tundra phytocenoses. All considered climatic scenarios include a slight NPP decrease for meadow steppes in European Russia (Fig. 6c). The IAP RAS CM and ECHAM4/OPYC3 models predict an insignificant NPP decrease for phytocenoses of dry steppes and semideserts in European Russia. According to the climatic scenario of ECHAM4/OPYC3, a decrease in NPP of phytocenoses in Siberian true steppes is probable (Fig. 6e). The HadCM3 scenario predicts favorable climatic conditions for the phytocenoses of true and dry steppes in Siberia.

Figures 7 and 8 show the proportion of the habitat where vegetation disappears, changes through succession, or remains unaltered for each considered type of zonal vegetation in Russia after climatic changes related to the increased mean annual global surface temperature by 1°C in the IAP RAS CM, ECHAM4/OPYC3, and HadCM3 models. According to the ECHAM4/OPYC3 and IAP RAS CM scenarios, the disappearance of the considered zonal phytocenoses is not expected. The HadCM3 climatic scenario includes an insignificant decrease in the habitat of *Pinus sylvestris* forests in both European Russia (by about 30000 km^2 for central and southern taiga forests in Upper Volga and the Volga–Vyatka Interfluve) and West Siberia (by about 21000 km^2: about 10000 km^2 for southern taiga forests and 11000 km^2 for steppe forests in the Upper Ob region).

Tundra phytocenoses whose habitats occupy about 270000 km^2 in European Russia and over 1380000 km^2 in Siberia and Far East will largely undergo no changes according to all considered climatic scenarios. Active successions are expected for these phytocenoses at an insignificant fraction of their habitats with the area of about 30000 km^2 in Asian Russia and from 6000 km^2 (HadCM3 scenario) to 21000–28000 km^2 (IAP RAS CM and ECHAM4/OPYC3 scenarios) in European Russia. The changes are expected in southern cottongrass–sedge tussock tundras of *Eriophorum vaginatum* and *Carex lugens* on the northeastern coast of the Shelikhov Bay and in the Anadyr–Velikaya Interfluve, southern shrub tundra of *Betula nana*, *B. exilis*, *Salix phylicifolia*, *S. glauca*, and *S. lanata* in the northeastern Kola Peninsula and Southeastern Yamal Peninsula. The IAP RAS CM and ECHAM4/OPYC3 scenarios predict possible changes in the phytocenoses of typical subshrub tundras with *Empetrum nigrum*, *S. herbacea*, and *Dryas punctata* in the southern Gydansk Peninsula.

The predicted climate changes will induce successions in most forest phytocenoses in Russia occupying over 2200000 km^2 in European Russia and over 5200000 km^2 of the plain areas in Siberia and Far East. According to the IAP RAS CM and ECHAM4/OPYC3 scenarios, the central and southern taiga *Pinus sylvestris* forests in European Russia are expected to change at a considerable part of the habitat—roughly from 260000 to 320000 km^2. The HadCM3 scenario predicts the changes in this phytocenosis only in the Vychegda–Kama Interfluve at the area no more than 28000 km^2. The ECHAM4/OPYC3 and IAP RAS CM scenarios predict the changes in the northern taiga *P. sylvestris* forests in southern Karelia on the area up to 40000 km^2. The climatic changes predicted by the considered scenarios will induce successions in *P. sylvestris* West Siberia forests: the changes are likely on the area of 40000–130000 km^2 in the northern taiga phytocenoses of this type; 130000–210000 km^2, in the southern taiga

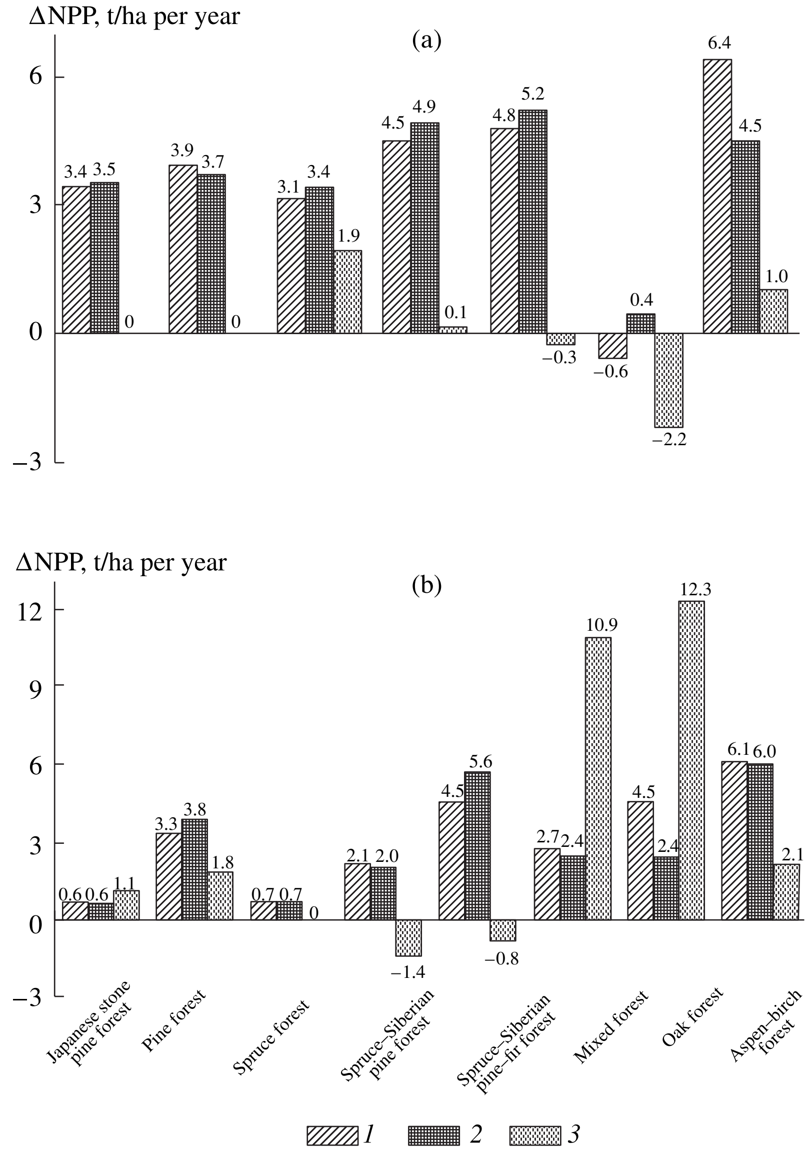

Fig. 6. Changes in the net primary production (ΔNPP) induced by climatic changes related to the increased mean annual global surface temperature by 1°C in the IAP RAS CM (*1*), ECHAM4/OPYC3 (*2*), and HadCM3 (*3*) models for forest vegetation types in European (a) and Asian (b) Russia and for grass vegetation types in European (c) and Asian (d) Russia.

phytocenoses; and 10000–30000 km^2, in the steppe phytocenoses. The IAP RAS CM and ECHAM4/OPYC3 scenarios predict successions in spruce forests in European Russia on the area of about 170000 km^2 for the central taiga *Picea abies* forests; 70000–100000 km^2 for the central taiga *P. obovata* forests; and 20000–100000 km^2 for the southern taiga *P. abies* forests. The ECHAM4/OPYC3 scenario also predicts changes in the northern taiga dark coniferous *P. abies* forests in the Onega–Northern Dvina Interfluve on the area of about 8000 km^2. For the considered climatic scenarios, successions are likely in the central taiga dark coniferous forests of *P. obovata* and *P. sibirica* in western Urals on the area from 10000 km^2 (HadCM3 prediction) to 30000 km^2 (ECHAM4/OPYC3 and IAP RAS CM predictions). According to the ECHAM4/OPYC3 and IAP RAS CM scenarios, the entire area of southern taiga dark coniferous forests of *P. obovata, P. sibirica*, and *Abies sibirica* in the western Urals (12000 km^2) will undergo successions. These climatic scenarios also assume the changes in the southern taiga dark coniferous forests of *P. obovata* and *A. sibirica* in the Vyatka–Kama Interfluve on the area of about 50000 km^2. Note that the HadCM scenario predicts no changes in the phytocenoses of spruce and spruce–Siberian pine–fir forests in European Russia. The ECHAM4/OPYC3 and IAP RAS CM scenarios expect the changes in the spruce–Siberian pine–fir forests of *P. obovata, P. sibirica*, and *A. sibirica* on considerable parts of their habi-

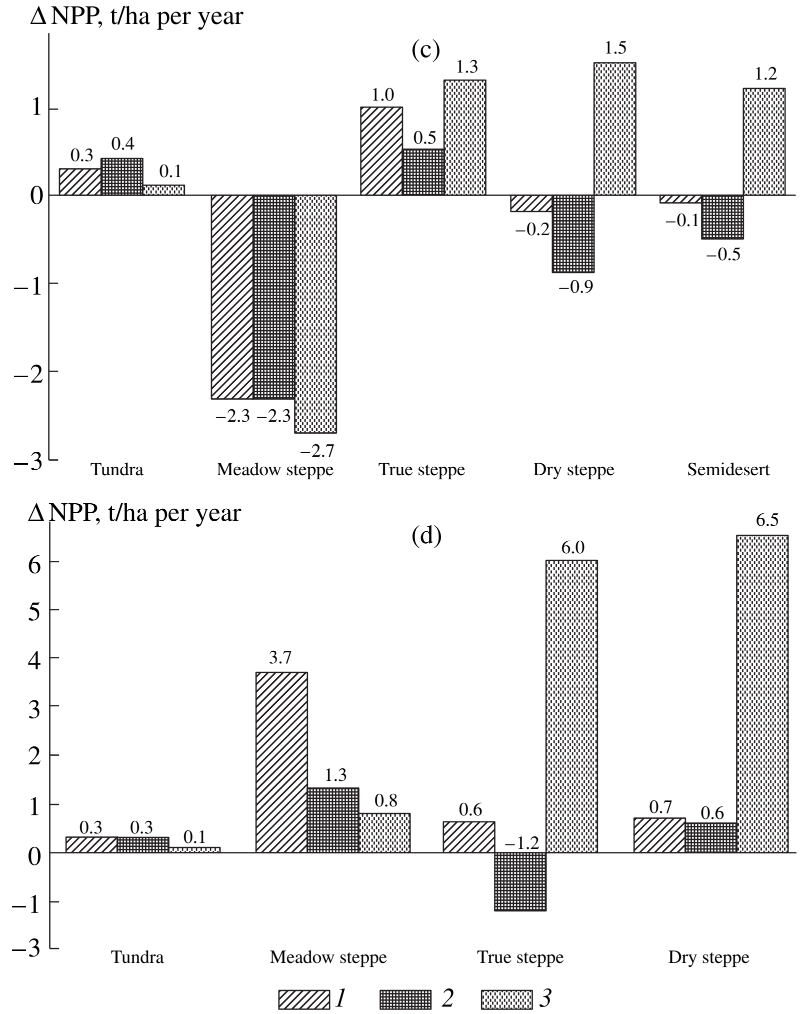

Fig. 6. Contd.

tats: 100000–200000 km^2 in the central taiga phytocenoses and 260000 km^2 in the southern taiga ones. The HadCM scenario predicts successions in these forest phytocenoses on the area of about 11000 and 70000 km^2, respectively. The climatic changes in the considered scenarios will induce successions on the area of 4000–17000 km^2 in the northern taiga larch and spruce–larch forests of *Larix sibirica* in West Siberia and on the area of 10000–60000 km^2 in the central taiga larch mouse–grass–subshrub and subshrub forests of *L. gmelinii.* According to the HadCM3 scenario, the larch paludal southern forest of *L. gmelinii* and *Sphagnum* sp. in Manchuria will undergo successions on the entire habitat of 52000 km^2, while the ECHAM4/OPYC3 and IAP RAS CM scenarios predict the changes in this phytocenoses on the area no more than 4000–6000 km^2. The ECHAM4/OPYC3 scenario expects the changes in the Central Siberian phytocenoses of both central and southern taiga larch forests of *L. sibirica* and *P. obovata* in the Yenisey–Stony Tunguska Interfluve as well as in the West Siberia phytocenoses of northern taiga larch–spruce–Siberian pine forests of *P. obovata, P. sibirica*, and *L. sibirica* in the Middle and Lower Vakh regions on a significant parts of their habitats (6000 and 10000 km^2, respectively). The HadCM3 scenario predicts the changes in the southern taiga larch forests of *L. gmelinii* on the area of about 15000 km^2 in Manchuria and larch paludal northern forest of *L. gmelinii* and *Sphagnum* sp. on the area of about 13000 km^2 in the Lower Amur region. The ECHAM4/OPYC3 and IAP RAS CM scenarios predict the changes in central taiga (on the area of about 25000 km^2) and southern taiga (on the area from 6000 to 35000 km^2) larch–pine Central Siberian forests of *L. sibirica* in the Yenisey–Angara Interfluve. The predicted climatic changes will induce successions in Japanese stone pine (*P. pumila*) and alder (*Duschekia kamtschatica* and *D. fruticosa*) forests occupying considerable areas in the plain part of the Okhotsk and Bering Coasts. Successions are likely in this phytocenoses in the northwestern Kamchatka Peninsula on the area of

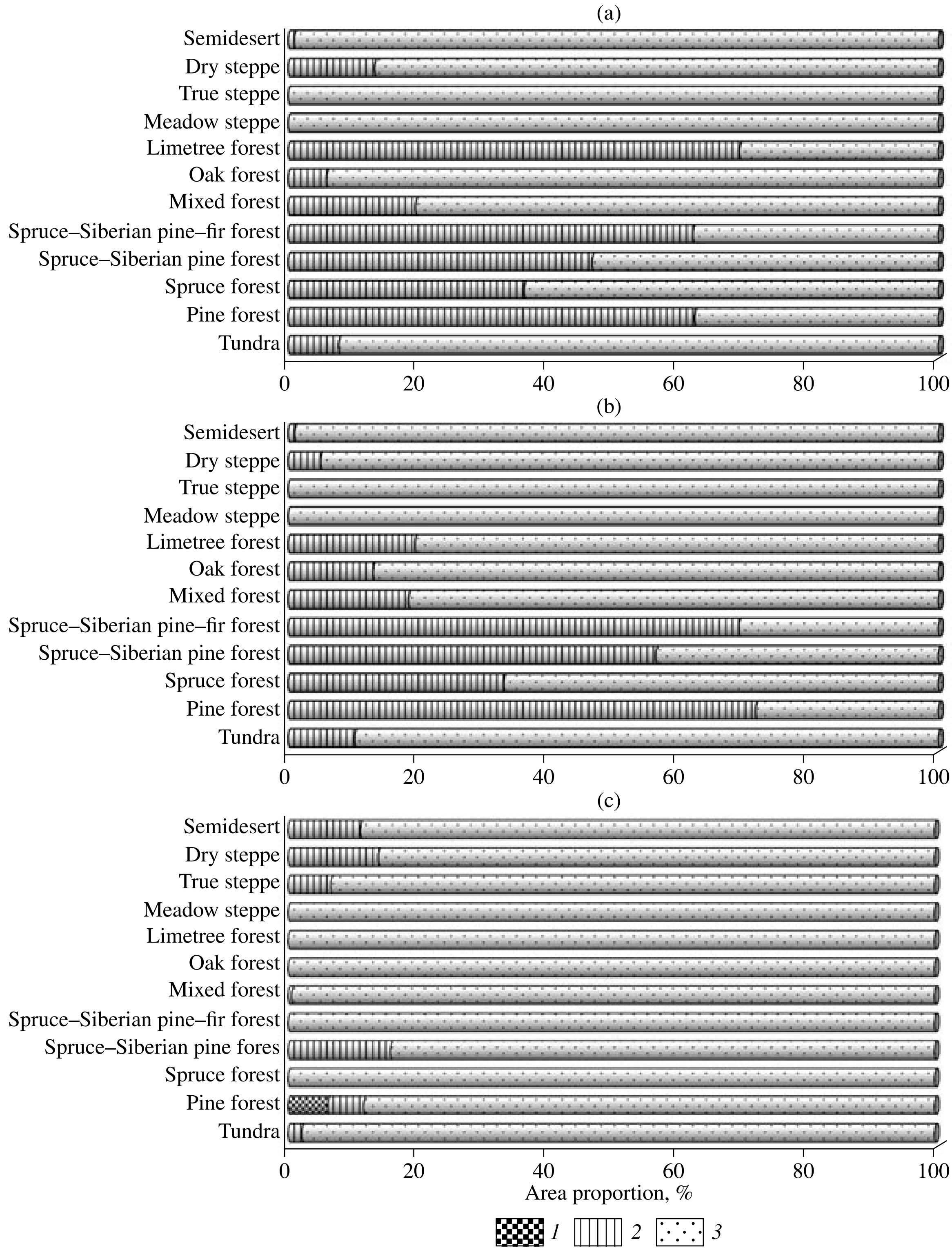

Fig. 7. Evaluated areas occupied by zonal vegetation types in European Russia where they vanish (*1*), undergo successions (*2*), or remain unaltered (*3*) after climatic changes related to the increased mean annual global surface temperature by 1°C in the IAP RAS CM (a), ECHAM4/OPYC3 (b), and HadCM3 (c) models.

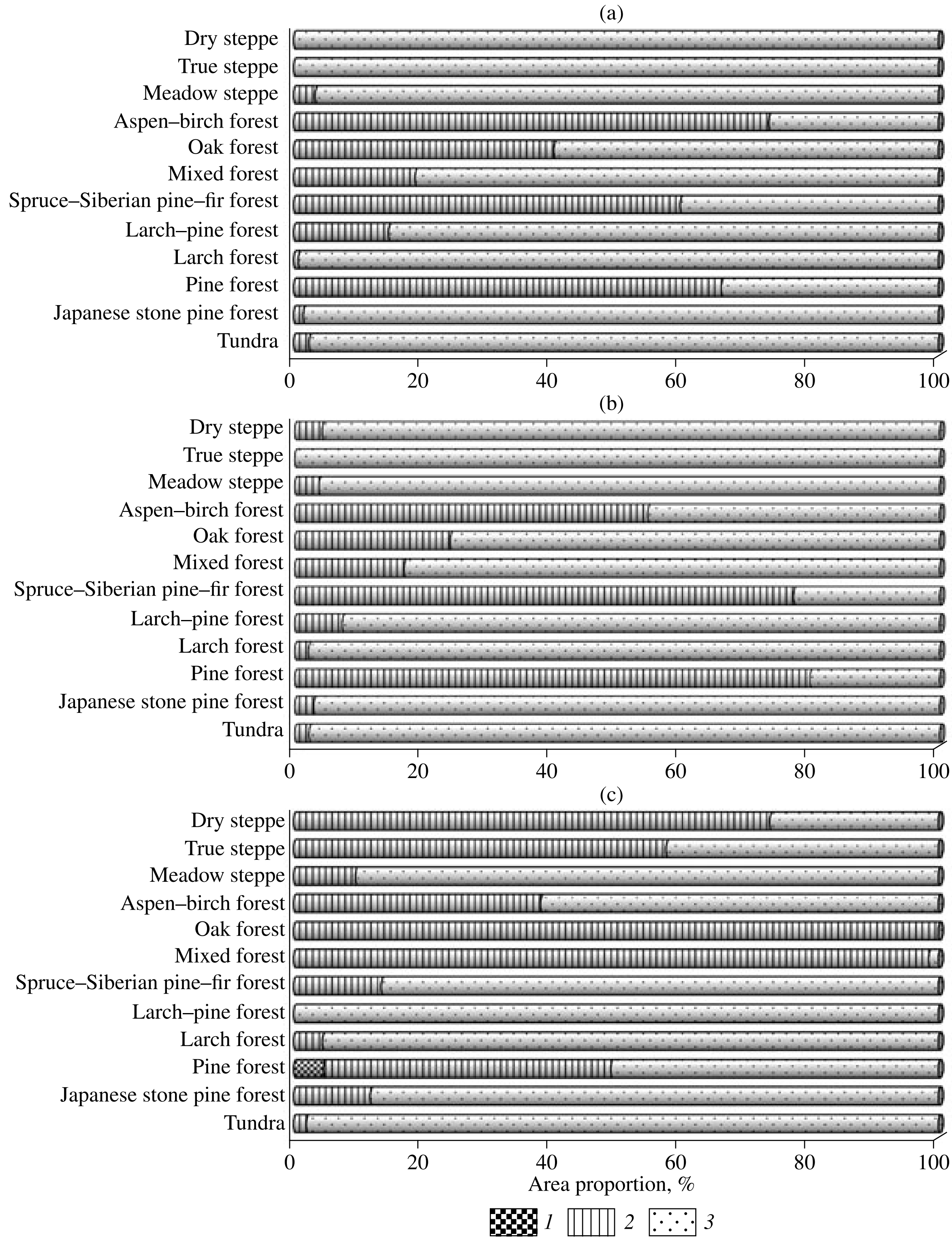

Fig. 8. Evaluated areas occupied by zonal vegetation types in Asian Russia where they vanish (*1*), undergo successions (*2*), or remain unaltered (*3*) after climatic changes related to the increased mean annual global surface temperature by 1°C in the IAP RAS CM (a), ECHAM4/OPYC3 (b), and HadCM3 (c) models.

about 5000 km^2 (according to all considered scenarios), in the southwestern Kamchatka Peninsula on the area of about 20000 km^2 (according to the HadCM3 scenario), and on the Bering Coast in the Lower Khatyrka and Pekul'neiskoe Lake regions on the area of about 10000 km^2 (according to the HadCM3 scenario). The considered climatic scenarios predict successions in broadleaf–coniferous forests of *P. obovata*, *A. sibirica*, and *Quercus robur* in the Middle Belaya region on the area from 2000 km^2 (according to the HadCM3 scenario) to 17000–25000 km^2 (according to the ECHAM4/OPYC3 and IAP RAS CM scenarios). The broadleaf–coniferous forests of *P. abies* and *Q. robur* in the Upper Volga region are predicted to undergo changes on the area of about 44000 km^2 according to the ECHAM4/OPYC3 and IAP RAS CM scenarios. The HadCM3 scenario predicts successions in the Siberian pine–broadleaf forests combined with spruce–fir forests of *P. koraiensis, A. holophylla*, and *A. nephrolepis* in Manchuria throughout the habitat (on the area of about 80000 km^2), while the ECHAM4/OPYC3 and IAP RAS CM scenarios predict similar changes for these forests on the area of about 22000 km^2. According to the HadCM3 scenario, successions will cover nearly all broadleaf–coniferous forests of *L. gmelinii* and *Q. mongolica* on the area of about 44000 km^2 in this region. The ECHAM4/OPYC3 and IAP RAS CM scenarios predict successions in limetree and oak–limetree forests of *Tilia cordata* and *Q. robur* at the foot of southern Urals on the area of 4000–13000 km^2 and in oak forests of *Q. robur* accompanied by other broadleaf forms (*T. cordata* and *Acer* spp.) in the eastern Smolensk–Moscow Upland and in the Meshchera Lowland on the area of 20000–40000 km^2. According to the HadCM3 scenario, all oak forests of *Q. mongolica* accompanied by *T. amurensis* on the area of about 55000 km^2 in Manchuria will undergo successions, while the ECHAM4/OPYC3 and IAP RAS CM scenarios predict successions in these forests on the area of about 13000 and 22000 km^2, respectively. Successions are likely in the aspen–birch forests of *B. pendula, B. pubescens*, and *Populus tremula* in West and Central Siberia. The changes in these forests are expected on the area from 64000 km^2 (HadCM3 scenario) to 90000–120000 km^2 (ECHAM4/OPYC3 and IAP RAS CM scenarios).

The expected climate warming will partially change the steppe phytocenoses occupying the area of about 628000 km^2 in European Russia and over 625000 km^2 in Siberia. All considered climatic scenarios suggest successions in the phytocenoses of cereal dry steppes with *Stipa lessingiana* in the southern Urals on the area from 6000 km^2 (ECHAM4/OPYC3 scenario) to 43000 km^2 (HadCM3 scenario). The HadCM3 scenario predicts changes in similar phytocenoses in the Kulunda steppe on the area of about 10000 km^2. The changes are expected in meadow steppes of *S. pennata, S. stenophylla*, and *Helictotrichon desertorum* and steppe meadows combined with birch and aspen outliers in the Irtysh–Ob Interfluve on the area of about 14000 km^2 (ECHAM4/OPYC3 and IAP RAS CM scenarios) and in the southern Urals and Ishim steppe on the area of about 36000 km^2 (HadCM3 scenario). The ECHAM4/OPYC3 and HadCM3 scenarios predict successions in the phytocenosis of cereals with *Cleistogenes squarrosa* in South Siberian dry steppes in Transbaikalia on the area from 4000 to 28000 km^2. According to the HadCM3 scenario, the changes are expected on the area of more than 110000 km^2 in the phytocenoses of forb–cereal semideserts with *S. lessingiana* and *S. zalesskii* in the Southern Urals and Kulunda steppe, forb–cereal true dry steppe with *S. lessingiana* and *S. zalesskii* in the Southern Urals, and true forb–cereal steppe with *Tanacetum sibirica* and *Leymus chinensis* in Transbaikalia.

Semidesert and desert phytocenoses in European Russia occupying the area of about 280000 km^2 will remain according to the considered climatic scenarios. The HadCM3 scenario predicts successions changes in the phytocenoses of sagebrush–cereal semidesert with *Artemisia lerchiana* and *A. sublessingiana* on the area of about 7000 km^2, cereal–sagebrush northern desert with *A. lerchiana, A. sublessingiana*, and *A. pauciflora* on the area of about 6000 km^2, and northern psammophytic desert with *Stipagrostis pennata, Koeleria glauca*, and *L. racemosus* on the area slightly more than 6000 km^2. The IAP RAS CM scenario predicts the changes in the phytocenosis of sagebrush–cereal semideserts with *A. lerchiana* and *A. sublessingiana* on an insignificant part of the habitat roughly 2000 km^2 in area.

All considered scenarios recognized the following phytocenoses sensitive to climatic changes: southern taiga forests of *P. obovata, P. sibirica*, and *A. sibirica*; central and southern taiga West Siberian forests of *P. sylvestris*; steppe pine Kazakhstan–West Siberian forests; West Siberian aspen–birch forests of *B. pendula, B. pubescens*, and *P. tremula*; and Far Eastern forests of *Q. mongolica* accompanied by *T. amurensis.* According to the ECHAM4/OPYC3 and HadCM3 scenarios, this group of phytocenoses also includes the central taiga dark coniferous forests of *P. abies* and *P. obovata*; southern taiga dark coniferous forests of *P. obovata* and *A. sibirica*; central and southern taiga forests of *P. sylvestris*; central taiga dark coniferous forests of *P. obovata* and *P. sibirica* in the western Urals region; southern taiga dark coniferous forests in the Western Urals region and central taiga forests in West Siberia of *P. obovata, P. sibirica*, and *A. sibirica*; northern taiga forests of *P. sylvestris* in West Siberia; sagebrush–cereal semideserts of *A. lerchiana* and *A. sublessingiana.* The IAP RAS CM scenario supplements this group with southern taiga dark coniferous forests of *P. abies* as well as limetree and oak–limetree forests of *T. cordata* and *Q. robur* at the foot of southern Urals.

According to the HadCM3 scenario, the phytocenoses sensitive to climatic impact also include larch paludal northern and southern forests of *L. gmelinii* and *Sphagnum* sp.; broadleaf–coniferous forests of *L. gmelinii* and *Q. mongolica*; pine–broadleaf forests of *Q. robur* and *P. sylvestris*; Far Eastern Siberian pine–broadleaf forests accompanied by spruce–fir forests of *P. koraiensis, A. holophyila*, and *A. nephrolepis*; sagebrush–bunchgrass dry steppes of *Artemisia maritima, A. taurica, S. capillata, Festuca valesiaca*, and *Agropyrum pectiniforme*; forb–cereal true moderately dry and true dry steppes of *S. lessingiana* and *S. zalesskii*; cereal dry steppes of *S. lessingiana*; East Siberian true forb–cereal steppes of *T. sibirica* and *L. chinensis*; cereal dry South Siberian steppes of *C. squarrosa*; and northern psammophytic desert with *Stipagrostis pennata, Koeleria glauca*, and *L. racemosus.*

CONCLUSIONS

The current climatic warming features an unprecedented rate of changes and spatiotemporal inhomogeneity. The distinctive features of modern warming predetermine a complex and ambiguous response of individual phytocenoses and ecosystems. Climate warming can cause considerable changes in the structure, species composition, and seasonal dynamics of plant communities, which should change the habitats of individual phytocenoses. In this work, possible changes in the habitats of zonal phytocenoses with regrowth vegetation for the plain territories of Russia were analyzed under a 1°C increase in the annual mean global surface temperature by using the climate models ECHAM4/OPYC3, HadCM3, and IAP RAS CM. The NPP changes for the considered climatic scenarios and the proposed rules of phytocenotic changes as a function of its NPP were used.

According to the HadCM3, the probable climate changes will reduce the habitat of central and southern taiga in European Russia by 6% and the habitats of southern taiga pine forests and steppe pine forest in West Siberia by about 4 and 23%, respectively.

The considered climatic scenarios predict the beginning or activation of successions in some part of the habitat of about a half of phytocenoses in Russia. Vegetation successions are expected on the area from 1000000 km^2 (HadCM3 scenario) to 2000000 km^2 (ECHAM4/OPYC3 scenario) in Russian plains with the total area of about 10000000 km^2. According to the IAP RAS CM and ECHAM4/OPYC3 scenarios of climatic changes, the entire southern taiga dark coniferous forests in western Urals region will undergo successions. The calculations by the HadCM3 model predict successions in the Far Eastern phytocenoses of larch paludal southern forests, oak forests accompanied by Amur linden, Siberian pine–broadleaf forests accompanied by spruce–fir forests throughout their habitats. According to this scenario, successions are probable in South Siberian cereal dry steppes in Transbaikalia and broadleaf–coniferous forests in the Far East on the bulk (over 70%) of their habitats. The IAP RAS CM and ECHAM4/OPYC3 scenarios predict successions in the phytocenoses of central taiga dark coniferous forests and central and southern taiga pine forests in European Russia, southern taiga spruce–Siberian pine–fir forests in West Siberia, central and southern taiga pine forests in West Siberia on the bulk (over 70%) of their habitats. Successions are likely in the northern taiga pine forests in West Siberia and aspen–birch forests in West Siberia on considerable parts of their habitats—over 80 and 73% (ECHAM4/OPYC3 and IAP RAS CM scenarios, respectively).

According to the considered scenarios of climate changes, the phytocenoses of meadow steppe in European Russia occupying the area of about 220000 km^2 will maintain their habitats despite the expected decreased in their NPP on average by 2–3 t/ha per year. The ECHAM4/OPYC3 and IAP RAS CM scenarios predict no significant changes in the phytocenoses of true steppe occupying in the area of about 288000 km^2 in European Russia and about 160000 km^2 in West and Central Siberia. The IAP RAS CM scenario predicts no changes in the phytocenoses of Siberian dry steppes occupying the area of about 937000 km^2. According to the ECHAM4/OPYC3 scenario, the phytocenoses of semideserts and deserts occupying the area of more than 207000 km^2 in southern European Russia will maintain their habitats.

According to the ECHAM4/OPYC3 and IAP RAS CM scenarios, all forest phytocenoses will more or less undergo successions in European Russia as well as in Siberia and Far East. The HadCM3 scenario predicts no changes in the following forest phytocenoses: spruce forests covering the area of more than 900000 km^2, spruce–Siberian pine–fir forests covering the area of about 93000 km^2, limetree forests covering the area of about 18000 km^2, and oak forests covering the area of more than 327000 km^2 in European Russia and larch–pine forests in the central and southern taiga covering the area of about 3400000 km^2 in Asian Russia.

The zonal phytocenoses of Russia sensitive to future climate changes in the second half of the 20th century occupied from 1100000 km^2 (HadCM3 scenario) to 2400000 km^2 (IAP RAS CM scenario).

The obtained data indicate a mosaic pattern of possible changes in the phytocenoses within their current habitats. The presented model estimates expose the spatial trends and scales of possible changes in the structure of Russian phytocenoses.

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